

SHETH L.U.J AND SIR M.V COLLEGE

SUBJECT: Artificial intelligence
PRACTICAL NO. 8

Aim: **Association Rule Mining**

- Implement the Association Rule Mining algorithm (e.g., Apriori) to find frequent itemsets.
- Generate association rules from the frequent itemsets and calculate their support and confidence.
- Interpret and analyze the discovered association rules.

CODE:

```
import pandas as pd
import matplotlib.pyplot as plt

from mlxtend.frequent_patterns import apriori, association_rules

# Load dataset
df = pd.read_csv("divorce.csv")

print("First five records:")
print(df.head())

# Select categorical and binary columns
columns = [
    "education_level",
    "employment_status",
    "religious_compatibility",
    "conflict_resolution_style",
    "marriage_type",
    "cultural_background_match",
    "mental_health_issues",
    "infidelity_occurred",
    "counseling_attended",
    "pre_marital_cohabitation",
    "domestic_violence_history",
    "divorced"
]
```

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```
data = df[columns].copy()

# Convert binary values to Yes/No
binary_columns = [
    "cultural_background_match",
    "mental_health_issues",
    "infidelity_occurred",
    "counseling_attended",
    "pre_marital_cohabitation",
    "domestic_violence_history",
    "divorced"
]

for column in binary_columns:
    data[column] = data[column].map({
        0: "No",
        1: "Yes"
    })

# Convert categorical data into transaction items
transactions = []

for _, row in data.iterrows():

    transaction = []

    for column in data.columns:
        transaction.append(
            column + "=" + str(row[column])
        )

    transactions.append(transaction)

# Get all unique items
all_items = sorted(
    set(
        item
        for transaction in transactions
        for item in transaction
    )
)
```

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)

Create transaction matrix

```
basket = pd.DataFrame(  
    False,  
    index=range(len(transactions)),  
    columns=all_items
```

)

for i, transaction in enumerate(transactions):

for item in transaction:

basket.loc[i, item] = True

print("\nTransaction Data:")

print(basket.head())

Apriori algorithm

```
frequent_itemsets = apriori(  
    basket,  
    min_support=0.20,  
    use_colnames=True
```

)

print("\nFrequent Itemsets:")

print(frequent_itemsets)

Generate association rules

```
rules = association_rules(  
    frequent_itemsets,  
    metric="confidence",  
    min_threshold=0.60
```

)

Remove invalid values

```
rules = rules.dropna(  
    subset=["support", "confidence", "lift"]
```

)

Sort by confidence

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```
rules = rules.sort_values(
    by="confidence",
    ascending=False
)

print("\nAssociation Rules:")

if len(rules) > 0:

    print(
        rules[
            [
                "antecedents",
                "consequents",
                "support",
                "confidence",
                "lift"
            ]
        ].head(10)
    )

    # Select top 5 rules for graph
    top_rules = rules.head(5).copy()

    # Create short rule names
    top_rules["rule"] = (
        top_rules["antecedents"].apply(
            lambda x: ", ".join(list(x))
        )
        + " -> " +
        top_rules["consequents"].apply(
            lambda x: ", ".join(list(x))
        )
    )

    print("\nTop Association Rules:")

    print(
        top_rules[
            [
```

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```
        "rule",
        "support",
        "confidence",
        "lift"
    ]
]
)

# Support and Confidence graph
plt.figure(figsize=(10, 6))

x = range(len(top_rules))

plt.bar(
    [i - 0.2 for i in x],
    top_rules["support"],
    width=0.4,
    label="Support"
)

plt.bar(
    [i + 0.2 for i in x],
    top_rules["confidence"],
    width=0.4,
    label="Confidence"
)

plt.xlabel("Association Rules")
plt.ylabel("Value")
plt.title("Support and Confidence")

plt.xticks(
    x,
    range(1, len(top_rules) + 1)
)

plt.legend()
plt.tight_layout()
plt.show()
```

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```
# Lift graph
```

```
plt.figure(figsize=(10, 6))
```

```
plt.bar(  
    range(len(top_rules)),  
    top_rules["lift"]  
)
```

```
plt.xlabel("Association Rules")  
plt.ylabel("Lift")  
plt.title("Lift of Association Rules")
```

```
plt.xticks(  
    range(len(top_rules)),  
    range(1, len(top_rules) + 1)  
)
```

```
plt.tight_layout()  
plt.show()
```

```
else:
```

```
print("No association rules found.")  
print("Try reducing the confidence threshold.")
```

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OUTPUT:

```
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First five records:
  age_at_marriage  marriage_duration_years  ...  trust_score  divorced
0                30                      1  ...    6.262411         1
1                27                      2  ...    6.769384         1
2                31                      6  ...    5.532866         1
3                35                      3  ...    3.491264         0
4                26                      2  ...   10.000000         1

[5 rows x 22 columns]

Transaction Data:
  conflict_resolution_style=Aggressive  ...  religious_compatibility=Same Religion
0                                     False  ...                               False
1                                     True  ...                               True
2                                     False  ...                               True
3                                     True  ...                               False
4                                     False  ...                               False

[5 rows x 29 columns]

Frequent Itemsets:
  support  itemsets
0  0.250000  frozenset({conflict_resolution_style=Aggressive})
1  0.583333  frozenset({conflict_resolution_style=Collabora...
2  0.916667  frozenset({counseling_attended=No})
3  0.333333  frozenset({cultural_background_match=No})
4  0.666667  frozenset({cultural_background_match=Yes})
...
1290 0.250000  frozenset({domestic_violence_history=No, divor...
1291 0.250000  frozenset({domestic_violence_history=No, menta...
1292 0.250000  frozenset({domestic_violence_history=No, marri...
1293 0.250000  frozenset({domestic_violence_history=No, divor...
1294 0.250000  frozenset({domestic_violence_history=No, cultu...

[1295 rows x 2 columns]

Ln: 30 Col: 65
```

```
IDLE Shell 3.13.2
File Edit Shell Debug Options Window Help

1293 0.250000  frozenset({domestic_violence_history=No, divor...
1294 0.250000  frozenset({domestic_violence_history=No, cultu...

[1295 rows x 2 columns]

Warning (from warnings module):
  File "C:\Users\gazala\AppData\Local\Programs\Python\Python313\Lib\site-packages\mlxtend\frequent_patterns\associ
  ation_rules.py", line 184
    cert_metric = np.where(certainty_denom == 0, 0, certainty_num / certainty_denom)
RuntimeWarning: invalid value encountered in divide

Association Rules:
  antecedents  ...  lift
16261  frozenset({religious_compatibility=Different Religion, em...  ...  3.000000
0      frozenset({conflict_resolution_style=Aggressive})  ...  1.090909
1      frozenset({conflict_resolution_style=Aggressive})  ...  1.500000
2      frozenset({conflict_resolution_style=Aggressive})  ...  1.000000
3      frozenset({conflict_resolution_style=Aggressive})  ...  1.500000
4      frozenset({conflict_resolution_style=Aggressive})  ...  1.200000
16245  frozenset({mental_health_issues=No, religious_...  ...  2.400000
16244  frozenset({mental_health_issues=No, religious_...  ...  3.000000
16238  frozenset({cultural_background_match=Yes, reli...  ...  2.400000
16237  frozenset({cultural_background_match=Yes, reli...  ...  2.000000

[10 rows x 5 columns]

Top Association Rules:
  rule  ...  lift
16261  religious_compatibility=Different Religion, em...  ...  3.000000
0      conflict_resolution_style=Aggressive -> counse...  ...  1.090909
1      conflict_resolution_style=Aggressive -> cultur...  ...  1.500000
2      conflict_resolution_style=Aggressive -> domest...  ...  1.000000
3      conflict_resolution_style=Aggressive -> marria...  ...  1.500000

[5 rows x 4 columns]

>>>

Ln: 30 Col: 65
```

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